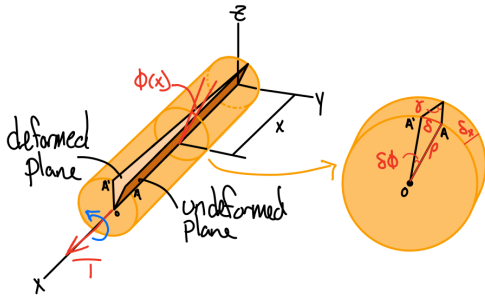


Torsional Deformation.

Induction. Angle of Twist



- ρ 当前 cross-section 半径
- c 总半径 最大
- γ 当前 cross-section Shear strain

$$\frac{d\phi}{dx} = \frac{\gamma}{\rho} = \frac{\gamma_{\max}}{c} \Rightarrow \gamma = \rho \frac{d\phi}{dx}$$
$$\tau = G\gamma$$

$$\frac{d\phi}{dx} = \frac{\tau}{JG} \Rightarrow \tau = \frac{Tp}{J}$$

- L : 总长度

$$\phi = \int_0^L \frac{\tau}{JG} dx = \frac{TL}{JG}$$

The Torsion Formula

$$\tau_{\max} = \frac{Tc}{J}$$

- T Torque
- c max radius (usually uniform)
- J polar Moment of Inertia.

$$J = \int_A \rho^2 dA$$

Solid Rod

$$J = \frac{\pi}{2} c^4$$

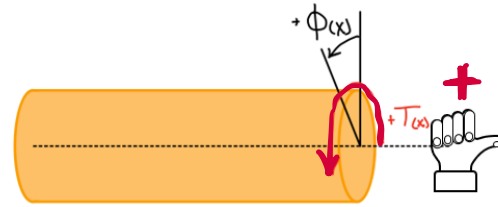
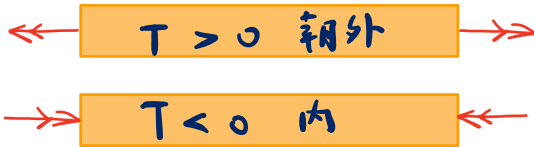
Hollow Tube

$$J = \frac{\pi}{2} (C_o^4 - C_i^4)$$

- C_o outer radius
 - C_i inner radius
- $$C_i + t = C_o$$

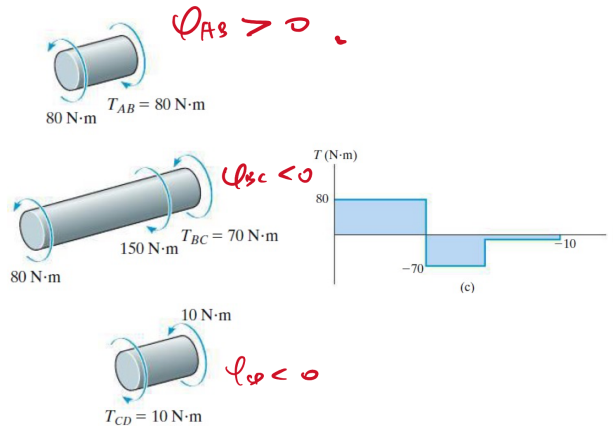
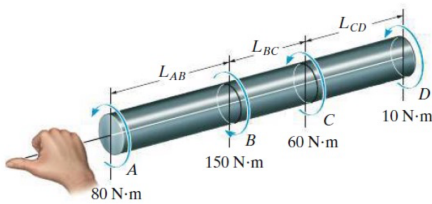
学会计算 $\min\{t\} \rightarrow$ $\star$ t thickness

Torque Sign Convention



- 角度分段算
- ① 认定一端正方向，一直参照这个
 - ② 外力 Torque 以此叠加 + 计算角度
 - ③ 同一个 rod，Torque 最终收束为 0
 - ④ 同一个 rod，角度 最终收束为 0

Sign Convention

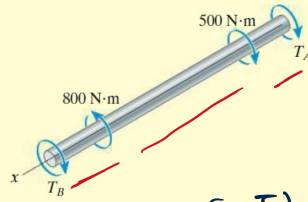
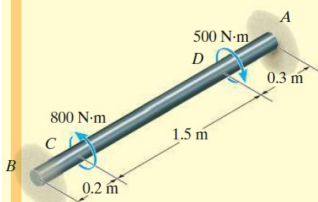


$$\phi_{A/D} = \frac{(+80 \text{ N}\cdot\text{m}) L_{AB}}{JG} + \frac{(-70 \text{ N}\cdot\text{m}) L_{BC}}{JG} + \frac{(-10 \text{ N}\cdot\text{m}) L_{CD}}{JG}$$

* 从一端算到另一端，一直叠加
 确定④后 角之和为0，不加符号

Example

The solid steel shaft shown in Fig. 5-23a has a diameter of 20 mm. If it is subjected to the two torques, determine the reactions at the fixed supports A and B.



① 读未知数，定方向

$T_A, T_B = \text{Symbols}$

② $-T_B + 800 - 500 - T_A = 0$

③ 角度总 **和** 为 0

$$0 = \frac{(-T_B) 0.2}{JG} + \frac{(-T_B + 800) 1.5}{JG} + \frac{(-T_B + 800 - 500) 1.5}{JG}$$

从一端算到另一端

Thin-walled Tube Analysis

Assume : linearly Distributed stress

$$\tau_{avg} = \frac{T}{2t A_m}$$

• t : thickness

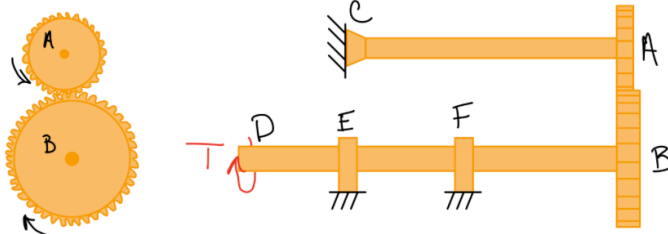
• A_m : Tube 中线 面积

$$A_m = \pi \left(\frac{R_i + R_o}{2} \right)^2$$

Gear Analysis

* Read Textbook Site
CAREFULLY.

Gear Systems



Gear system

Extra!

Power transmission can be used to find shear stress in gear shafts.

Maximum shear stress:

$$\tau = \frac{Pc}{\omega J}$$

Gear system problems can be solved using the torsion equations.

Common assumptions:

- Gears are perfectly rigid.
- The rotation axis is perfectly fixed in space.
- Gear teeth are evenly spaced and perfectly shaped so there is no gap to create lost motion.
- Gear tooth faces are perfectly smooth so there is no slip.
- Mated gears twist through the same arc length.

To solve:

1. Draw a FBD for each shaft.
2. Use the torsion equilibrium equations of each shaft to find the reaction forces and torques.
3. Use the shear stress equation to find the stresses in the shaft.

The angle of twist between two mating gears is related through the gear ratio.

Gear ratio.

$$GR = d_B : d_A$$

Angle of twist

$$\phi_A = GR \phi_B$$